FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 1.1: PRESSURE | LESSON 9: FINAL EXPLANATION & SYNTHESIS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

You have now completed all 9 lessons on Pressure. This is your moment to bring everything together. Below, you will find six phenomena — the same ones you investigated across this unit, starting with the crushed jerrycan that launched your curiosity on Day 1. For EACH phenomenon, write a complete scientific explanation using the guiding prompts. A strong explanation names the phenomenon clearly, identifies the pressure concept(s) at work, states the correct formula(s), links to evidence from your lessons, and uses logical scientific reasoning to connect cause and effect.

HOW TO USE THIS DOCUMENT:

- Read the phenomenon description carefully before writing.
- Use the bullet-point prompts inside each section — they tell you exactly what to include.
- Reference your Summary Table, lab notes, and Driving Question Board entries.
- Write in full sentences. Bullet lists alone are not enough.
- After completing all six phenomena, complete the Synthesis section to show how all the concepts connect.

SELF-CHECK BEFORE SUBMITTING: Ask yourself —

✔️ Did I name the correct formula for every phenomenon?
✔️ Did I explain WHY the formula matters in this situation (not just write it down)?
✔️ Did I include at least one piece of evidence from a lesson experiment or observation?
✔️ Did I connect my explanation back to the crushed jerrycan or another real Kenyan context?
✔️ Did I write in clear, complete sentences that someone outside this class could understand?

Phenomenon 1 — The Crushed Jerrycan (Anchoring Phenomenon)	
The Phenomenon: Your teacher sealed an empty 20-litre plastic jerrycan — the kind used every day across Kenya to carry	The jerrycan crushed itself because of atmospheric pressure — the pressure exerted by the mass of air in Earth's atmosphere pressing down and inward on every surface below it. At sea level, atmospheric pressure is approximately 101,325 Pa (about 101,325 newtons pushing on every square metre of surface). This pressure acts in ALL directions — upward, downward, and sideways — not just downward like gravity.

water from boreholes, rivers, and water points. A small amount of air was removed from inside. Slowly, without anyone touching it, the jerrycan crumpled and collapsed inward on all sides. The same jerrycan routinely carries 20 kg of water without deforming. Why did removing a little air cause it to crush itself? **Write your complete explanation. Make sure you include ALL of the following:** **→ What atmospheric pressure is and where it comes from (hint: think about what air is made of and how much of it sits above you).** **→ The formula P = F/A and what each symbol means.** **→ What was happening INSIDE the jerrycan before and after air was removed.** **→ Why the pressure outside became greater than the pressure inside — and what that imbalance caused.** **→ Why the same thing does NOT crush your body (even though atmospheric pressure is pushing on you right now).** **→ Evidence from Lesson 4 or Lesson 5 that supports your explanation.** **→ A real Kenyan example or connection (e.g., water-carrying jerrycans in your county, vacuum-sealed packaging of tea from Kericho).**	Using the formula P = F/A, where P is pressure (Pa), F is force (N), and A is the contact area (m^2), we can understand what happened. When the jerrycan was full of air, the air pressure INSIDE matched the atmospheric pressure OUTSIDE. The forces were balanced — outside pressure pushing inward was equal to inside pressure pushing outward — so the jerrycan kept its shape. When a small amount of air was removed, the number of air molecules inside decreased. Fewer molecules means fewer collisions against the inner walls, which means LOWER pressure inside. But the atmospheric pressure outside stayed the same — roughly 101,325 Pa — still pushing powerfully inward on every face of the jerrycan. Now the forces were no longer balanced: outside pressure (high) > inside pressure (low). This net inward force had nothing to oppose it, so the walls of the jerrycan were pushed inward — it crushed itself. The reason our bodies do not get crushed is that our bodies contain fluids and gases (blood, air in our lungs, fluid in our tissues) that push outward with the same pressure as the atmosphere pushes inward. The pressures are balanced, so we feel nothing unusual — just as the jerrycan was fine when it was full of air. Evidence from Lesson 4 supports this: in the inverted glass experiment, we saw that atmospheric pressure was strong enough to hold up a full column of water against gravity — proving that this invisible force is enormously powerful. In Lesson 5, we learned that atmospheric pressure decreases with altitude because there is less air above us at higher elevations, confirming that the weight of air above is what creates this pressure. This is also why vacuum-sealed packets of Kericho Gold tea are compressed tightly — the air inside is removed, and atmospheric pressure outside squeezes the packet inward, just like the jerrycan.

Phenomenon 2 — Sharp Panga vs. Blunt Panga

The Phenomenon: A farmer in Kisumu uses the same arm strength to chop sugarcane with a sharp panga and then with a blunt panga. The sharp panga slices through easily; the blunt panga barely makes a dent. The force applied is identical. Why does the sharpness of the blade make such a dramatic difference? **Write your complete explanation. Make sure you include ALL of the following:** → **The formula P = F/A and what each symbol means.** → **How the contact area (A) of a sharp blade compares to a blunt blade.** → **What happens to pressure (P) when area decreases but force stays the same — and vice versa.** → **Why high pressure at the blade edge is needed to cut through material.** → **Evidence from the clay/pencil experiment in Lesson 1 or calculations from Lesson 2.** → **At least TWO other real-life Kenyan examples that use the same principle (think: needles at a clinic, nails in carpentry, high heels on soft ground, thorns on acacia trees).**	This phenomenon is explained by the pressure formula P = F/A, where P is pressure in pascals (Pa), F is the applied force in newtons (N), and A is the contact area in square metres (m^2). The formula tells us that for the same force, a SMALLER contact area produces a HIGHER pressure. A sharp panga has a very thin, fine edge — its contact area (A) with the sugarcane is extremely small, perhaps just a fraction of a square centimetre. A blunt panga has a wide, flat edge — its contact area is much larger. When the farmer applies the same force (F) with both pangas, the sharp panga concentrates that force into a tiny area, creating very high pressure at the point of contact. This high pressure is great enough to overcome the internal forces holding the sugarcane fibres together, so the blade cuts through. The blunt panga spreads the same force over a large area, creating much lower pressure — not enough to break through the fibres, so it bruises rather than cuts. In our Lesson 1 clay experiment, we pressed a sharp pencil tip and a blunt eraser end into clay using the same hand force. The sharp pencil tip sank deeply into the clay while the eraser barely left a mark. This is identical in principle to the panga: same force, different area, very different pressure. Lesson 2 calculations confirmed this numerically — doubling the area halves the pressure for the same force. Two other Kenyan examples of this principle: 1. Clinic needles in Nairobi's hospitals have an extremely fine tip (tiny A), so a small push from a nurse creates enough pressure to pierce skin painlessly and quickly. A blunt needle would require enormous force for the same effect. 2. A woman wearing narrow high-heeled shoes at a graduation ceremony in Mombasa sinks into soft ground because all her weight is concentrated on a tiny heel area, creating very high pressure — much higher than flat sandals would produce with the same body weight.

Phenomenon 3 — Why the Gitaru Dam Wall is Thicker at the Bottom	
The Phenomenon: Engineers who designed the Gitaru hydroelectric dam on the Tana River (and other dams like	This dam design is explained by the formula for pressure in liquids: P = ρgh, where P is the pressure at a given point (Pa), ρ (rho) is the density of water (approximately 1,000 kg/m^3), g is the gravitational field strength (approximately 10 N/kg), and h is the depth below the water surface (m). This formula tells us three things: pressure in a liquid increases as density increases, as gravity increases, and — most importantly here — as DEPTH increases.

Seven Forks along Kenya's rivers) made the walls much thicker at the bottom than at the top. This uses more concrete and costs more money. Why is this necessary? Why can't the dam wall be the same thickness all the way down? **Write your complete explanation. Make sure you include ALL of the following:** **→ The formula P = ρgh and what each symbol means (ρ = density of water, g = gravitational field strength, h = depth below the water surface).** **→ How pressure in a liquid changes with depth — and WHY (think about the weight of water above any given point).** **→ What the pressure is like near the TOP of the dam vs. near the BOTTOM, using the formula.** **→ Why the dam wall must be thickest where pressure is greatest.** **→ Evidence from the Lesson 3 water-bottle experiment (holes at different heights).** **→ A connection to any other structure or context in Kenya (e.g., water storage tanks in Nairobi, fish at depth in Lake Victoria).**	The reason pressure increases with depth is that the deeper you go, the greater the weight of water sitting above that point. Each extra metre of water adds another column of weight pressing down and outward. At the very top surface of the dam reservoir, h = 0, so the pressure from the water alone is zero (only atmospheric pressure acts). At the bottom of a 40-metre-deep reservoir, P = 1,000 × 10 × 40 = 400,000 Pa — four hundred thousand pascals pushing outward against the dam wall. The pressure at the bottom is enormously greater than at the top. Because the water pressure pushing outward on the dam wall is highest at the bottom, the wall must be thickest there to provide enough strength to resist that force without cracking or collapsing. Making the top equally thick would waste expensive concrete without any engineering benefit. In our Lesson 3 experiment, we made holes at the top, middle, and bottom of a plastic water bottle. Water shot out farthest from the bottom hole and barely trickled from the top hole — directly demonstrating that pressure (and therefore the force pushing water outward) is greatest at the greatest depth. This is exactly what the dam engineers must account for. Fish living deep in Lake Victoria experience this same increasing pressure with depth. And the large water storage tanks on elevated stands across Nairobi must have strong base connections precisely because the water pressure at the outlet pipe (at the base) is greater than anywhere else in the tank.

Phenomenon 4 — The Hydraulic Jack at a Nairobi Garage	
The Phenomenon: At a matatu garage along Landhies Road in Nairobi, a mechanic uses a hydraulic jack to lift a 14-seater matatu that weighs thousands	This phenomenon is explained by Pascal's Principle, which states: pressure applied to a confined, incompressible fluid is transmitted equally and without loss in all directions throughout the fluid. The hydraulic jack is a machine built entirely around this principle. The relevant formula is $F_1/A_1 = F_2/A_2$, which can also be written as $P_1 = P_2$. Here, F_1 is the force the mechanic applies to the small

of newtons. He pushes down on the jack handle with a modest, manageable force — and the heavy vehicle rises smoothly off the ground. How can such a small human push lift such an enormous load? **Write your complete explanation. Make sure you include ALL of the following:** **→ Pascal's Principle — state it clearly in your own words.** **→ The formula $F_1/A_1 = F_2/A_2$ (or equivalently $P_1 = P_2$) and what each symbol represents.** **→ How the small piston and large piston work together — trace the sequence of events from the mechanic's push to the matatu rising.** **→ Why the output force (F_2) is so much larger than the input force (F_1).** **→ Evidence from the connected-syringes experiment in Lesson 6.** **→ At least ONE other hydraulic system used in Kenya (e.g., hydraulic brakes on buses, excavators at construction sites in Nairobi, hospital dental chairs).**	(input) piston, A_1 is the area of that small piston, F_2 is the force produced at the large (output) piston, and A_2 is the area of that large piston. Because the hydraulic oil connecting both pistons is incompressible, the pressure created at the small piston is transmitted perfectly to the large piston. Here is the sequence of events: The mechanic pushes down on the jack handle with force F_1 on the small piston (small area A_1). This creates a pressure $P = F_1/A_1$ in the hydraulic oil. By Pascal's Principle, this same pressure P acts on the large piston (large area A_2). Therefore the force pushing up on the large piston is $F_2 = P \times A_2$. Since A_2 is much greater than A_1, F_2 is much greater than F_1. The matatu rises. For example: if $A_2 = 50 \times A_1$, then $F_2 = 50 \times F_1$. A mechanic pushing with just 200 N could lift 10,000 N — enough to raise a loaded matatu. In our Lesson 6 connected-syringes experiment, pushing the plunger of a small syringe caused the plunger of a large syringe to move outward with noticeably greater force — we directly experienced Pascal's Principle and force multiplication with our own hands. Another critical Kenyan application: hydraulic braking systems on long-distance buses travelling from Nairobi to Mombasa along the Mombasa Highway. When the driver presses the brake pedal, that small foot force is transmitted through brake fluid to large brake callipers at each wheel, generating the enormous clamping force needed to safely slow a fully loaded bus. Without Pascal's Principle, this would be impossible.

Phenomenon 5 — Ears Popping on Mount Kenya	
The Phenomenon: A group of students from Nyeri travelling to climb Mount Kenya notice that as they ascend above 3,000 metres, their ears feel increasingly blocked and uncomfortable. When they swallow or yawn, their ears	Both the blocked ears and the puffed crisp packet are caused by the decrease in atmospheric pressure with altitude. Atmospheric pressure exists because the weight of all the air in the atmosphere above us presses down on every surface. The higher you climb, the less air there is above you, so the weight pressing down — and therefore the pressure — decreases. At sea level (for example, at the Nyeri tuck shop), atmospheric pressure is approximately 101,325 Pa. The air pressure inside the ear (in the middle ear cavity) equals the pressure outside, so the eardrum sits in its natural, comfortable position — balanced forces on both sides. As the students ascend Mount Kenya, the outside atmospheric pressure drops steadily. But the air inside the middle ear cannot escape quickly — it remains at the higher pressure from lower altitude. Now the inside pressure is GREATER

make a 'pop' sound and suddenly feel clear again. Some students also notice that a sealed packet of crisps they brought from the tuck shop at the base has puffed up and become tightly inflated at high altitude. What is happening — and why? **Write your complete explanation. Make sure you include ALL of the following:** **→ What atmospheric pressure is and how it changes with altitude (use the idea of the weight of air above you).** **→ What is happening INSIDE the ear at sea level vs. at high altitude — and why the imbalance causes discomfort.** **→ Why swallowing or yawning causes the 'pop' and relieves the discomfort.** **→ Why the crisp packet puffed up at high altitude (connect to the jerrycan phenomenon — same concept, opposite situation).** **→ Evidence from Lesson 5 activities (personal experience accounts, barometer readings, or altitude-pressure graph).** **→ One practical Kenyan application: why aeroplane cabins flying between Nairobi and Mombasa are pressurised.**	than the outside pressure. This pressure imbalance pushes the eardrum outward, stretching it — causing the feeling of blockage and discomfort. The Eustachian tube connects the middle ear to the back of the throat. When a student swallows or yawns, this tube momentarily opens, allowing a small puff of air to escape from the middle ear. The inside pressure drops to match the outside pressure — the eardrum returns to its natural position — and the student hears and feels the 'pop' as the pressures equalise. The crisp packet puffed up for the opposite reason: the air sealed inside the packet at sea level has a higher pressure than the reduced atmospheric pressure at high altitude. The excess inside pressure pushes the packet walls outward — the packet inflates. This is the exact reverse of the crushed jerrycan: in the jerrycan, inside pressure was LESS than outside, so it was crushed inward; in the crisp packet, inside pressure is MORE than outside, so it is pushed outward. In Lesson 5, our class barometer readings and altitude-pressure data showed clearly that pressure falls as altitude rises — confirming all of the above reasoning. This is also why passenger aircraft flying between Wilson Airport in Nairobi and Moi International Airport in Mombasa maintain pressurised cabins: the air at cruising altitude is far too thin (low pressure) for humans to breathe comfortably, so the cabin is kept at an artificially higher pressure equivalent to a low altitude, protecting passengers' ears, lungs, and comfort.

Phenomenon 6 — Camels at Marsabit, Humans at the Beach	
The Phenomenon: Camels used by pastoralists in Marsabit County carry heavy loads and walk easily across deep, loose	This surprising result is perfectly explained by the pressure formula $P = F/A$, where P is the pressure exerted on the sand (Pa), F is the weight (force) pressing down (N), and A is the contact area of the foot on the sand (m^2). Pressure is not just about how heavy something is — it depends equally on how that weight is spread across an area.

sand without sinking. A person standing on the same sand sinks in up to their ankles, even though they weigh far less than a camel. Why does the heavier animal sink less than the lighter human? **Write your complete explanation. Make sure you include ALL of the following:** **→ The formula P = F/A and what each symbol means.** **→ How the structure of a camel's foot (wide, flat, padded) produces lower pressure than a human foot — even when the camel is heavier.** **→ A sample calculation to prove this numerically (use: Human — weight 600 N, foot area 0.04 m²; Camel — weight 5,000 N, foot area 0.8 m²).** **→ The design principle: to reduce sinking or damage, INCREASE contact area.** **→ Evidence from Lesson 8 calculations.** **→ THREE other Kenyan examples that use the same design principle (think about: wide tyres on lorries carrying maize from the Rift Valley, jua kali welders using wide base plates, elephants' wide feet in Tsavo National Park, foundation design for buildings in Nairobi).**	A camel's feet are large, wide, and padded — specially adapted for desert terrain. A human foot is comparatively small and narrow. Let us use the sample calculation from Lesson 8 to prove the point: Human: $P = F/A = 600\ N \div 0.04\ m^2 = 15{,}000\ Pa$ Camel: $P = F/A = 5{,}000\ N \div 0.8\ m^2 = 6{,}250\ Pa$ The camel weighs more than eight times as much as the person, yet it exerts LESS pressure on the sand — only 6,250 Pa compared to the human's 15,000 Pa. Because pressure (not raw weight) determines how deeply something sinks into a soft surface, the camel sinks less. Its wide feet distribute its enormous weight across a large area, keeping the pressure low enough that the sand can support it. The design principle is clear: if you want to avoid sinking into a soft surface, increase your contact area to reduce pressure. Three Kenyan examples of this principle in action: 1. Lorries transporting maize harvests from Nakuru and the Rift Valley to Nairobi's grain stores use wide, dual rear tyres. This spreads the lorry's heavy load over a larger road area, reducing pressure on the tarmac and preventing the road from cracking or the vehicle from sinking on murram roads after rain. 2. Elephants in Tsavo National Park have enormous, round, flat feet — a natural adaptation that spreads their 4,000–6,000 kg body weight across a large area, allowing them to walk quietly and without sinking into soft riverbank mud. 3. Construction engineers building tall apartments in Nairobi (such as in Kilimani or Westlands) design wide concrete foundation slabs under the building. The wide base spreads the entire building's enormous weight over a large ground area, reducing the pressure on the soil enough to prevent the building from sinking unevenly — a phenomenon called subsidence.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Phenomenon Description & Conceptual Accuracy	Clearly and accurately describes the phenomenon in own words, correctly	Describes the phenomenon adequately and identifies the correct pressure	Describes the phenomenon vaguely or inaccurately. Identifies only one relevant

	identifies ALL relevant pressure concept(s) (P=F/A, P=ρgh, Pascal's Principle, atmospheric pressure), and explains the causal mechanism with no misconceptions. All formulas are stated and every symbol is defined.	concept(s) with minor gaps. Formulas are present but one or two symbols may be undefined or one causal link may be incomplete.	concept or uses an incorrect formula. Causal reasoning is weak or missing, and symbols are largely undefined.
Use of Evidence from Lessons & Experiments	Cites specific, accurate evidence from at least one named lesson experiment or observation for each phenomenon (e.g., 'In the Lesson 3 water-bottle experiment, the bottom hole shot water farthest'). Evidence is clearly connected to the explanation and strengthens the argument.	References lesson experiments for most phenomena, but the connection between evidence and explanation is sometimes vague (e.g., 'we did an experiment that showed this'). At least one phenomenon lacks a specific evidence citation.	Evidence is rarely cited or is too vague to be meaningful (e.g., 'we saw it in class'). Two or more phenomena have no supporting evidence from lessons.
Scientific Reasoning & Cause-Effect Logic	Explanation traces a clear, logical cause-effect chain for every phenomenon (e.g., 'air removed → fewer molecules → lower inside pressure → imbalance → net inward force → collapse'). Reasoning is complete, sequential, and leaves no logical gaps. Distinguishes correctly between force and pressure throughout.	Cause-effect reasoning is mostly clear and correct but may skip one step in the chain or conflate force and pressure in one instance. The overall logic is followable and mostly sound.	Reasoning is incomplete, circular, or relies on everyday language without scientific mechanism (e.g., 'the air was stronger so it pushed'). Multiple logical gaps are present, or force and pressure are consistently confused.
Real-World Kenyan Connections & Applications	Each phenomenon explanation includes at least one specific, accurate Kenyan context (named locations, tools, animals, industries, or people). The Synthesis section lists multiple diverse applications across all four pressure domains (solids, liquids, atmospheric, Pascal's). Connections are meaningful and enhance understanding.	Most phenomena include a Kenyan context but one or two are generic or not specific to Kenya. The Synthesis section covers applications but may miss one pressure domain.	Kenyan contexts are absent or superficial in most phenomena (e.g., just writing 'in Kenya' without a real example). The Synthesis section lists fewer than three applications or omits entire pressure domains.
Synthesis: Connecting All Pressure Concepts	The Synthesis section clearly and accurately explains ALL four pressure domains (P=F/A in solids, P=ρgh in liquids, atmospheric pressure, Pascal's Principle), describes how they are related, and reflects genuine insight about how pressure is a unifying concept across the unit. References the crushed jerrycan as the anchoring phenomenon and shows how it connects to other phenomena.	The Synthesis section addresses at least three of the four pressure domains and makes a reasonable attempt at connecting them. The jerrycan may be mentioned but the connection to other phenomena is partial.	The Synthesis section is a simple list with little connection between concepts, or covers fewer than two pressure domains. No meaningful reflection on the unit as a whole is present, and the anchoring phenomenon is not mentioned.